Evaluating Normality and Constant Variance

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Math 269 - Statistical Consulting
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I. INTRODUCTION

A. Why is normality good?

- Satisfy the assumption to run many of the statistical procedures including regression, t tests, analysis of variance, and other parametric tests.
- Central limit theorem
- The sample mean and sample variance are independent ⇔ The errors in estimating the mean is completely unrelated to that in estimating the variance.
- If A and B are two variables from normal distributions then A + B and A - B are both normally distributed.

I. INTRODUCTION

A. What if normality is violated?[1]

- Technically, in regression, normality is not required if all we care about is estimating the slope coefficients.
- However it is required in order to make inference about the model. Normality violation may lead to:
  - P-values for significance tests may be inaccurate and lead to wrong conclusions.
  - Confidence intervals and prediction intervals can be too wide or too narrow.

I. INTRODUCTION

B. Why is Constant Variance good?

- **In Regression:** Part of the underlying assumption is that the error terms are iid normal with constant variance.
  - This essentially means that the response Y has variance σ^2 that doesn't depend on x.
- **In Time Series:** Part of the assumption of stationarity is that the variance of the process doesn't change over time.
  - Similar to regression, but x is *time.*
- **In ANOVA:** Assumption is that all the different groups have the same variance.
  - In this case the x is the different levels of the treatments.

I. INTRODUCTION

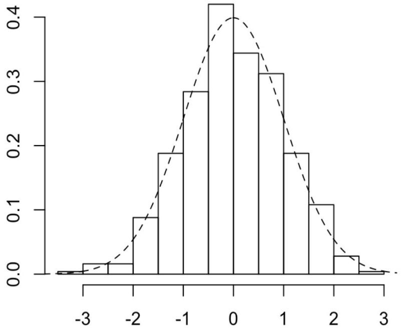
B. What if Constant Variance is violated?[1]

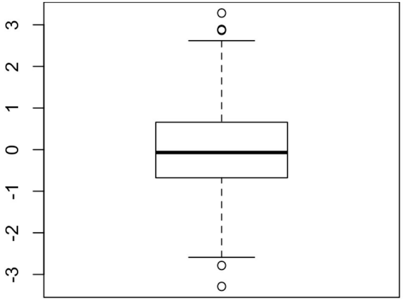
- Difficult to estimate the true value of the variance of the errors.
    - Confidence intervals can be too wide or too narrow.
- Difficult estimating coefficients because the subset of data with higher variance could have disproportionately more weight on the model estimates than the data with lower variance.

II. NORMALITY EVALUATION

A. Graphical methods:

1. Naive Methods: Histogram and Box-Plot





- Good at showing if a distribution is approximately *symmetric*, but hard to judge normality in particular.
- Histogram is very sensitive to the choice of bin width.
    - Can possibly lead to very misleading conclusions on its own.
- Box plot is better at showing kurtosis/the length of the tails, but it’s still hard to judge normality visually without using some other plot or formal hypothesis test.

II. NORMALITY EVALUATION

A. Graphical methods:

2. Q-Q Plot and P-P Plot

- Quantile-quantile plot (a.k.a. Q-Q Plot) plots the theoretical quantiles from a normal distribution against the empirical quantiles from the sample.
- Probability plot (a.k.a. P-P Plot) is very similar, but instead of using quantiles, it uses values from the normal CDF. Area under the curve to the left of a given point.
- Q-Q plots are more commonly used in practice.
- In general, Q-Q plot is better at finding deviations in the tails of the distribution and P-P plot is better at finding deviations in the middle of the distribution.[2]

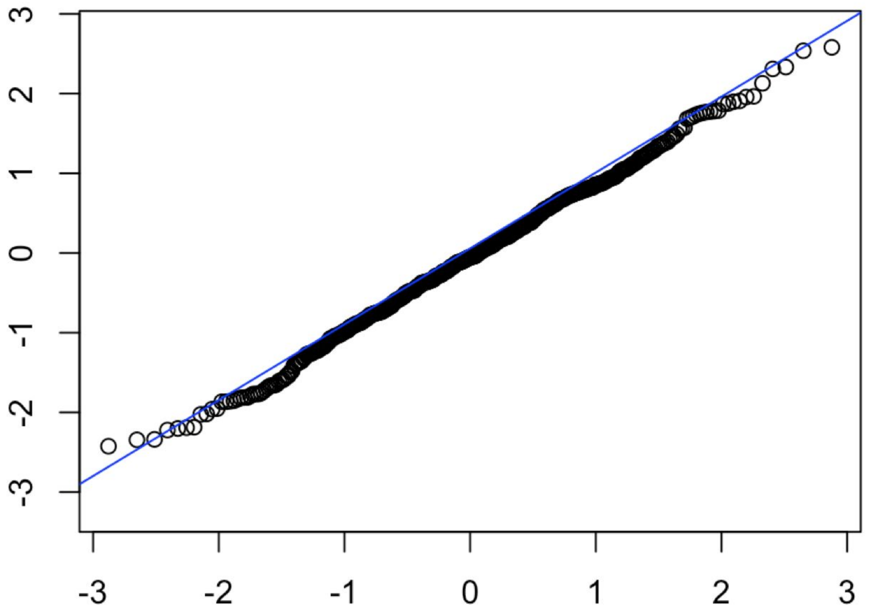
II. NORMALITY EVALUATION

A. Graphical methods:

2. Q-Q Plot

Example:

```
sample1 = rnorm(500)
x = seq(0,1,length.out = length(sample1))
q = qnorm(x,0,1)
plot(q,sort(sample1))
abline(0,1,col="blue")
```



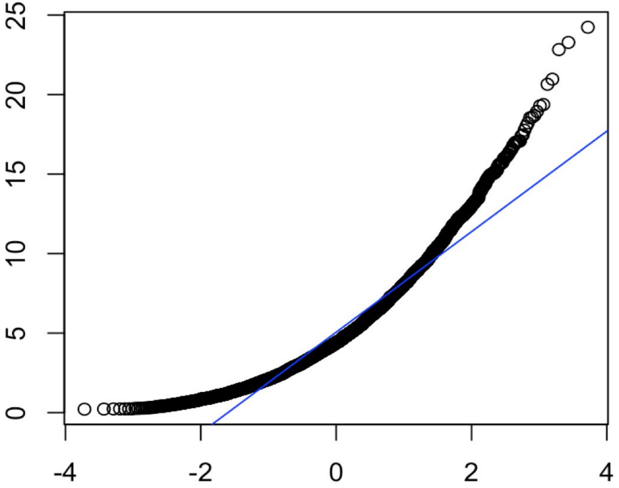
II. NORMALITY EVALUATION

A. Graphical methods:

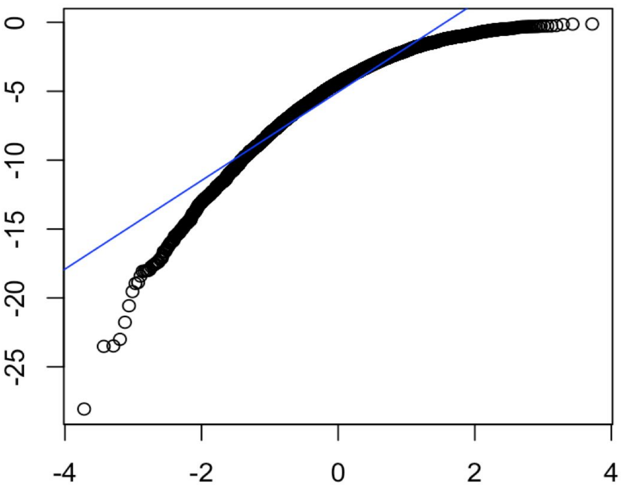
2. Q-Q Plot

Example:

Skewed right



Skewed left



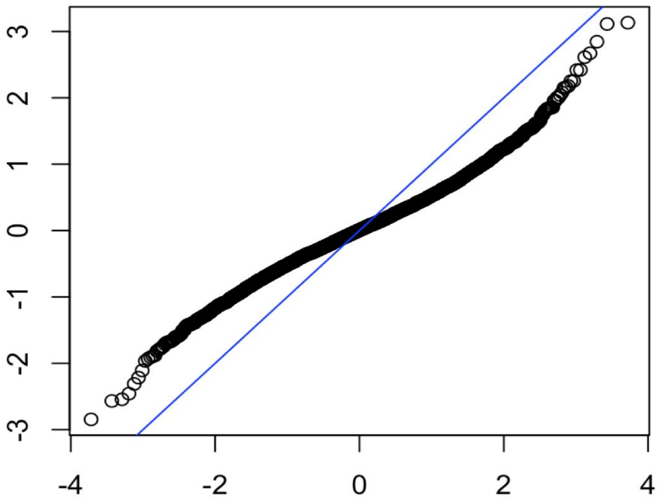
II. NORMALITY EVALUATION

A. Graphical methods:

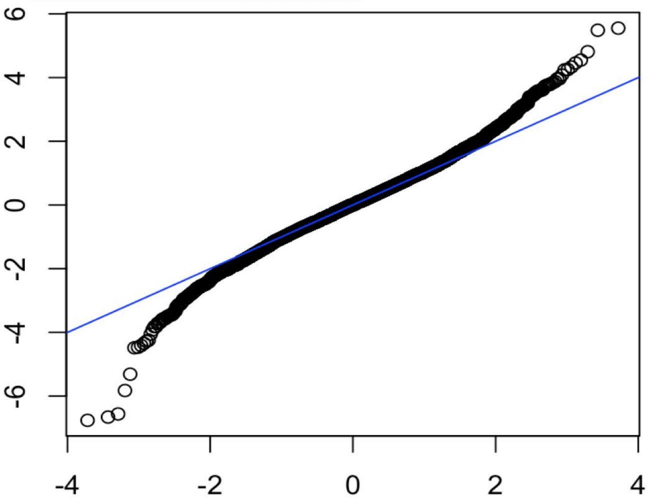
2. Q-Q Plot

Example:

Thin tails



Thick tails



II. NORMALITY EVALUATION

A. Graphical methods:

3. In Higher Dimensions: Chi-Square Q-Q Plot

- If $x_1, x_2, \dots, x_n$ is a random sample from a p-dimensional multivariate normal distribution, then $d_1^2, d_2^2, \dots, d_n^2$ where ${d_i}^2 = (\boldsymbol{x_i} - \boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x_i} - \boldsymbol{\mu})$ is a random sample from a χ_p^2 distribution.[3]
- Any formal goodness of fit test to test if $\boldsymbol{d}^2$ came from a chi-square distribution can test the hypothesis that $\boldsymbol{x}$ came from a multivariate normal distribution.
- Graphically, we can make a Q-Q plot of $d_1^2, d_2^2, \dots, d_n^2$ against the quantiles of a χ_p^2 distribution.

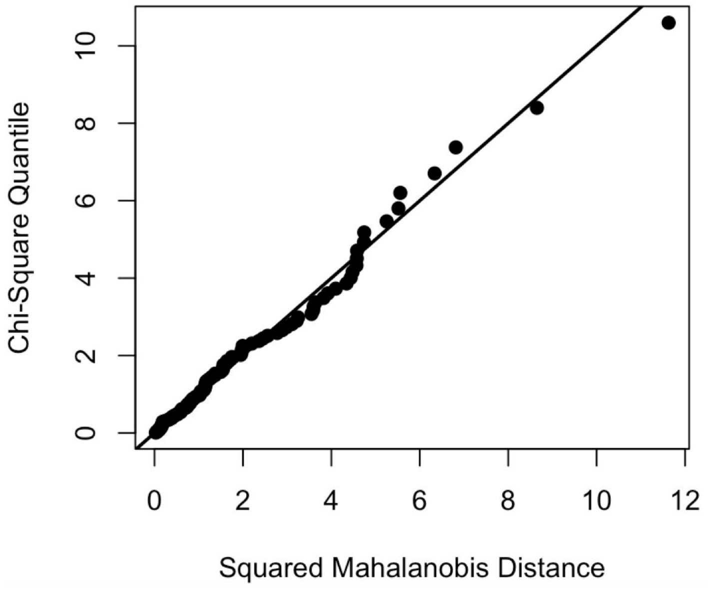
II. NORMALITY EVALUATION

A. Graphical methods:

3. Multivariate Normality: Chi-Square Q-Q Plot

<u>Example:</u>

```
library(MASS)

library(MVN)

X = mvrnorm(100, mu = c(0,10),

Sigma = matrix(c(10,1,1,3),nrow=2))

mvn(X, multivariatePlot = "qq")
```



II. NORMALITY EVALUATION

B. Statistical tests

1. Shapiro-Wilk test

- First published in 1965 by Samuel Sanford Shapiro and Martin Wilk.[4]
- Compares the observed data to where we would expect to see the data based on normal quantiles, much like a qq-plot.
- Assumptions:
  - The data is continuous and iid.
- Hypothesis Test:
  - H_0: The sample $(x_1, x_2, \dots, x_n)$ came from a Normally distributed population.
  - H_a: The sample $(x_1, x_2, \dots, x_n)$ did not come from a Normally distributed population.

II. NORMALITY EVALUATION

B. Statistical tests

1. Shapiro-Wilk test

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- Test Statistic:
  - $x_{(i)}$ is the i^{th} order statistic.
  - $a = \frac{m^T V^{-1}}{(m^T V^{-1} V^{-1} m)^{1/2}}$
    - m is a vector of expected values of the order statistics from a true Normal Distribution.
    - V is the covariance matrix of m
- Both numerator and denominator are approximating the variance of the population, so if the null hypothesis is true, W will be equal to 1.
- Small values of W are evidence against the null hypothesis.

II. NORMALITY EVALUATION

B. Statistical tests

1. Shapiro-Wilk test

- A table of critical values for W was published by Pearson and Hartley in 1972.
  - https://www.epa.gov/sites/production/files/2015-10/documents/monitoring_appendd_1997.pdf
- The test is unreliable when the sample size is very large. It can detect very slight deviations from what it expects to see and is likely to produce a type 1 error.
  - R won't even let you use a sample size larger than 5000.

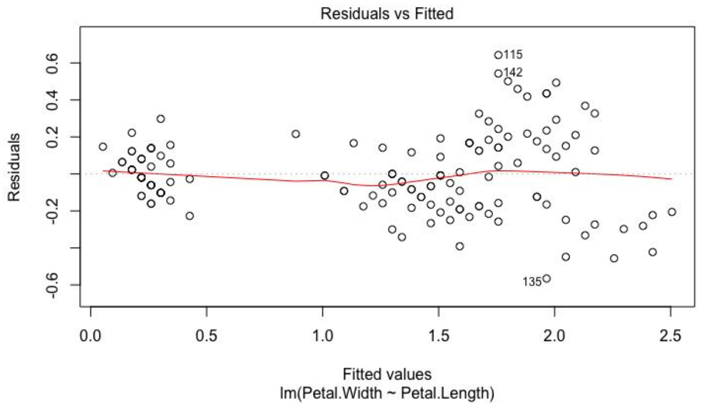
Example:

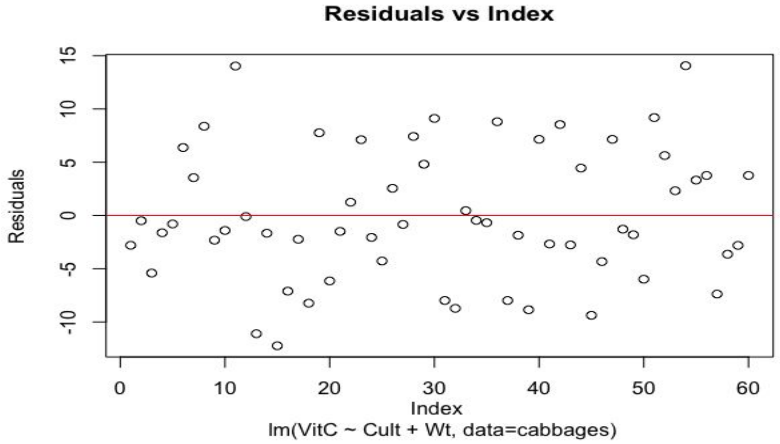
```
> sample1 = runif(100)
> shapiro.test(sample1)

        Shapiro-Wilk normality test

data:  sample1
W = 0.95106, p-value = 0.0009752
> sample2 = rnorm(100)
> shapiro.test(sample2)

        Shapiro-Wilk normality test

data:  sample2
W = 0.98607, p-value = 0.378
```

III. CONSTANT VARIANCE EVALUATION

A. Graphical methods:

1. Scatter plot





III. CONSTANT VARIANCE EVALUATION

A. Graphical methods:

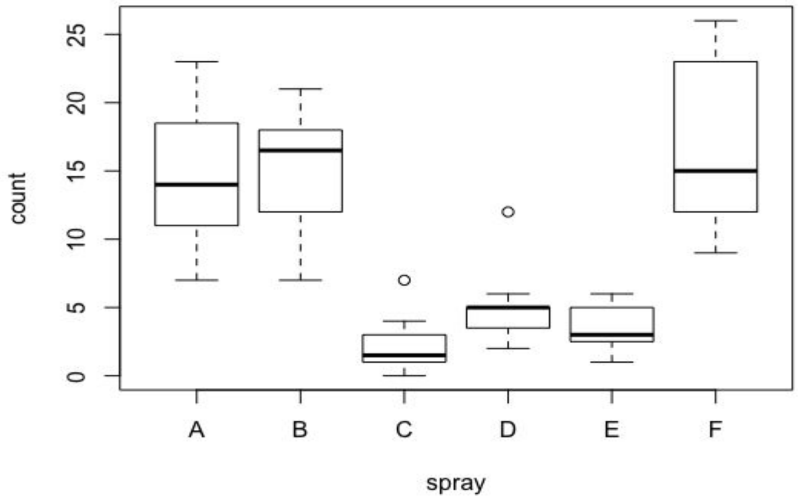
2. Box-plot

Dataset: InsectSprays

Name: Effectiveness of Insect Sprays

Description: The counts of insects in agricultural experimental units treated with different insecticides.

Format: A data frame with 72 observations on 2 variables “insect count” and “the type of spray” (6 factors: A to F)



III. CONSTANT VARIANCE EVALUATION
B. Statistical tests

- ❑ The residuals is divided into different groups
- ❑ The number of groups $g \geq 2$ with the size $n_1, n_2, \ldots, n_g$ such that their sum adds up to n
- ❑ Sample variance of group i:

$$s_i^2 = \frac{\sum_{j=1}^{n_i} (e_{i,j} - \bar{e}_{i,.})^2}{n_i - 1},$$

where ei,j is the jth residual from group i

- ❑ Pool variance:

$$s_p^2 = \frac{\sum_{i=1}^{g} (n_i - 1) s_i^2}{n - g}.$$

- ❑ Hypothesis test:

$$H_0\colon \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$$

$$H_a\colon \sigma_i^2 \neq \sigma_j^2 \text{ for at least one pair of } i, j$$

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

1. F- test [5]

Partition the residuals of observations into two groups

One consisting of residuals associated with the lowest predictor values and the other consisting of those belonging to the highest predictor values.

Assumption:

- The larger variance should always be placed in the numerator
- The population from which the samples were obtained come from normal distribution
- The samples are independent

Hypothesis test:

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_A : \sigma_1^2 \neq \sigma_2^2$$

Reject Ho when $F = s_1^2 / s_2^2 \geq F_{n_1-1,n_2-1;1-\alpha}$

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

2. Bartlett's test [6]

Assumption:

The samples follow normal distribution

Test statistic:

$$B = \frac{(n-g)\ln s_p^2 - \sum_{i=1}^{g}(n_i - 1)\ln s_i^2}{1 + \left(\frac{1}{3(g-1)}\left(\left(\sum_{i=1}^{g}\frac{1}{n_i - 1}\right) - \frac{1}{n-g}\right)\right)}$$

Under Ho, B ~ χ^2_{g-1}

Reject Ho if $B \geq \chi^2_{g-1;1-\alpha}$

III. CONSTANT VARIANCE EVALUATION
B. Statistical tests

2. Bartlett's test

Example: Revisit the Effectiveness of Insect Sprays

```
> bartlett.test(count ~ spray, data = InsectSprays)

	Bartlett test of homogeneity of variances

data:  count by spray
Bartlett's K-squared = 25.96, df = 5, p-value = 9.085e-05
```

The formula of the form lhs ~ rhs in the code where lhs gives the data values and rhs are the corresponding groups.

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

2. Bartlett's test

Higher dimensions:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{12} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1p} & \sigma_{2p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Assumption:

The samples are from multivariate normal distributions

$n_j - p$ is large for all j = 1, 2, … ,k (where k = # of groups, p = # of dependent variables)

Hypothesis test:

$$H_0 : \Sigma_1 = \Sigma_2 = \cdots = \Sigma_k$$

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

2. Bartlett's test

Test statistic:

$$M = \sum_{j=1}^{k}(n_j - 1)\ln|S_{pool}| - \sum_{j=1}^{k}(n_j - 1)\ln|S_j|$$

|Spool| = the determinant of the pooled estimate of the covariance matrix

|Sj| = the determinant of the covariance matrix for the jth group

Under the assumptions, $MC^{-1} \sim \chi^2_{df} \quad df = \frac{1}{2}(k-1)(p+1)p$

Where

$$C^{-1} = 1 - \frac{2p^2 + 3p - 1}{6(p+1)(k-1)}\left(\sum_j \frac{1}{n_j - 1} - \frac{1}{\sum_j (n_j - 1)}\right)$$

Reject Ho if $MC^{-1} \geq \chi^2_{df}(\alpha)$

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

2. Bartlett's test

Example [7]:

n1 = 45 (# Wisconsin homeowners **with** air conditioning)

n2 = 55 (**without** air conditioning)

X1 = a measure of total **on-peak** consumption during July (electrical usage in kilowatt hours)

X2 = **off-peak**

Summary statistics:

$$\bar{\mathbf{x}}_1 = \begin{bmatrix} 204.4 \\ 556.6 \end{bmatrix}, \quad \mathbf{S}_1 = \begin{bmatrix} 13825.3 & 23823.4 \\ 23823.4 & 73107.4 \end{bmatrix}, \quad n_1 = 45$$

$$\bar{\mathbf{x}}_2 = \begin{bmatrix} 130.0 \\ 355.0 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 8632.0 & 19616.7 \\ 19616.7 & 55964.5 \end{bmatrix}, \quad n_2 = 55$$

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

2. Bartlett's test

Example [7]:

```
> nac <- 45
> nnac <- 55
> n <- nac+nnac
> k <- 2
> p <- 2
>
> xbarac <- c(204.4,556.6)
> xbarnac <- c(130,355)
>
> mu0 <- c(0,0)
> dac <- (xbarac - xbarnac)
>
> Sac <- matrix(c(13825.3,23823.4,23823.4,73107.4), 2, 2, byrow = T)
> Snac <- matrix(c(8632, 19616.7,19616.7,55964.5), 2, 2, byrow = T)
> Spool <- (nac-1)/(n-2)*Sac + (nnac-1)/(n-2)*Snac
```

III. CONSTANT VARIANCE EVALUATION
B. Statistical tests
2. Bartlett's test

Example [7]:

```
> ## Bartlett's test
> (M <- (nac-1)*log(det(Spool))+(nnac-1)*log(det(Spool))-(nac-1)*log
(det(Sac))-(nnac-1)*log(det(Snac)))
[1] 19.36726
> (Cinv <- 1-((2*p^2+3*p-1)/(6*(p+1)*(k-1)))*(1/(nac-1)+1/(nnac-1)
-1/(n-2)))
[1] 0.977581
> df <- 1/2*(k-1)*(p+1)*p
>
> M*Cinv > qchisq(0.05,df,lower.tail = FALSE)
[1] TRUE
```

Reject Ho if $MC^{-1} \geq \chi^2_{df}(\alpha)$

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

3. Levene's test

- First proposed by Howard Levene in 1960.[8]
- Alternative to Bartlett's test, which is more sensitive to departures from non-normality.
- Assumptions:
  - independence within groups and between groups.
- Hypothesis Test:
  - H_0: $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$
  - H_a: $\sigma_i^2 \neq \sigma_j^2$ for at least one pair of i, j.

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

3. Levene's test

- Test Statistic:

$$W = \frac{N-k}{K-1} \frac{\sum_{i=1}^{k} n_i(\bar{Z}_{i.}-\bar{Z}_{..})^2}{\sum_{i=1}^{k} \sum_{j=1}^{n_i} (Z_{ij}-\bar{Z}_{i.})^2}$$

- $Z_{ij} = |X_{ij} - \bar{X}_{i.}|$
  - $\bar{X}_{i.}$ is either the mean, median, or trimmed mean of the i^{th} subgroup.
  - Median and trimmed mean are more robust against deviations from normality.
  - Median is default in R.
- Under the null hypothesis, $W \sim F(k-1, N-k)$.
- Large value of W is evidence against the null hypothesis.

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

3. Levene's test

Cautions:

Levene's original proposal of centering at the sample mean, has the correct Type I error rate only for symmetric distributions, while centering at the sample median has correct Type I error rate both for symmetric and for asymmetric distributions.[9]

Example:

```
> library("car")

> sample1 = rnorm(10)
> sample2 = rnorm(10)
> sample3 = rnorm(10)

> ( df = data.frame(y=c(sample1,sample2,sample3),
group=rep(c("a","b","c"),each=length(sample1))) )

> leveneTest(df$y, df$group, center = "median")

Levene's Test for Homogeneity of Variance (center = "median")
      Df F value Pr(>F)
group  2  0.6617 0.5241
      27
```

	y	group
1	-0.431583423	a
2	0.356758636	a
3	-1.062695576	a
4	1.145757386	a
5	-0.312309055	a
6	0.644917982	a
7	0.718844080	a
8	-1.747197766	a
9	-1.527560557	a
10	-0.674059744	a
11	1.026344481	b
12	-1.021511027	b
13	-0.356198018	b
14	-0.827645768	b
15	0.837505272	b
16	0.564685952	b
17	0.987572450	b
18	1.351465065	b
19	1.518981673	b
20	-1.910482579	b
21	0.844396103	c
22	0.491903859	c
23	1.132674466	c
24	0.736098795	c
25	0.516380635	c
26	-0.636676416	c
27	1.051573388	c
28	0.181529914	c
29	-1.622377035	c
30	-0.007237894	c

III. CONSTANT VARIANCE EVALUATION
B. Statistical tests

4. Brown-Forsythe test

a.k.a. modified Levene test

- Proposed by Morton B. Brown and Alan B. Forsythe in 1974.[10]
- The Levene test uses deviations from group **means**

-> a highly-skewed set of data -> may violate the assumption of normality

- The Brown-Forsythe correct this skewness by using deviations from group **medians**.

Assumption:

- ❑ Errors need not come from normal distribution
- ❑ The data are obtained by taking a simple random sample from each of the g populations

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

4. Brown-Forsythe test

Test statistic:

When g = 2,

$$\boxed{L = \frac{\bar{d}_1 - \bar{d}_2}{s_L\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}}$$

where $d_{i,j} = |e_{i,j} - \tilde{e_{i,\cdot}}|$, where $\tilde{e_{i,\cdot}}$ is the median of the i^{th} group of residuals.

$$s_L = \sqrt{\frac{\sum_{j=1}^{n_1}(d_{1,j} - \bar{d}_1)^2 + \sum_{j=1}^{n_2}(d_{2,j} - \bar{d}_2)^2}{n_1 + n_2 - 2}}$$

Under the null hypothesis, $L \sim t_{n1+n2-2}$ or $L^2 \sim F_{1,\ n1+n2-2}$ approximately.

III. CONSTANT VARIANCE EVALUATION

B. Statistical tests

4. Brown-Forsythe test

Cautions:

Glass and Hopkins (1996 p. 436) state that the Levene and B-F tests are “fatally flawed”; It isn’t clear how robust they are when there are significant differences in variances and unequal sample sizes.[11]

Example:

```
install.packages("onewaytests")
library(onewaytests)
out <- bf.test(Sepal.Length ~ Species, data = iris)

  Brown-Forsythe Test (alpha = 0.05)
-------------------------------------------------------------
  data : Sepal.Length and Species

  statistic  : 119.2645
  num df     : 2
  denom df   : 123.9255
  p.value    : 1.317059e-29

  Result     : Difference is statistically significant.
-------------------------------------------------------------

> paircomp(out)

  Bonferroni Correction (alpha = 0.05)
-----------------------------------------------------
   Level (a)  Level (b)       p.value   No difference
1     setosa versicolor 1.124023e-16          Reject
2     setosa  virginica 1.190060e-24          Reject
3 versicolor  virginica 5.598433e-07          Reject
-----------------------------------------------------
```

References

[1] http://people.duke.edu/~rnau/testing.htm

[2] https://www.theanalysisfactor.com/anatomy-of-a-normal-probability-plot/

[3] http://users.stat.umn.edu/~kb/classes/5401/files/ChisqQQPlot.pdf

[4] Shapiro, S. S., and M. B. Wilk. 1965. "An Analysis of Variance Test for Normality (Complete Samples)." Biometrika 52 (3-4): 591–611. https://www.jstor.org/stable/pdf/2333709.pdf

[5] https://www.itl.nist.gov/div898/handbook/eda/section3/eda359.htm

[6] Snedecor, George W. and Cochran, William G. 1989. "Statistical Methods, Eighth Edition." Iowa State University Press.

[7] Example from Richard A. Johnson and Dean W. Wichern. 2008. "Applied multivariate statistical analysis, 6th ed." Pearson. The R code is produced by Dr. Andrea Gottlieb from the course Math 257 at SJSU.

References

[8] Levene, H. 1960. “Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling.” Stanford University Press, pp. 278-292.

[9] Nordstokke, D.W. & Zumbo, B.D. 2007. "A Cautionary Tale about Levene's Tests for Equal Variances." Journal of Educational Research & Policy Studies. 7(1), 1-14. https://www.learntechlib.org/p/62682/

[10] Brown, Morton B.; Forsythe, Alan B. (1974). "Robust tests for the equality of variances". Journal of the American Statistical Association. 69: 364–367. https://www.jstor.org/stable/pdf/2285659.pdf

[11] Glass, Gene V., and Kenneth D. Hopkins. 1996. “Statistical Methods in Education and Psychology.” Boston, MA: Allyn and Bacon. p 436.

R Documentation

- https://www.rdocumentation.org/packages/graphics/versions/3.6.2/topics/hist
- https://www.rdocumentation.org/packages/graphics/versions/3.6.2/topics/boxplot
- https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/Normal
- https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/qqnorm
- https://www.rdocumentation.org/packages/MASS/versions/7.3-51.5/topics/mvrnorm
- https://www.rdocumentation.org/packages/MVN/versions/5.8/topics/mvn
- https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/shapiro.test
- https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/bartlett.test
- https://www.rdocumentation.org/packages/car/versions/3.0-6/topics/leveneTest
- https://www.rdocumentation.org/packages/onewaytests/versions/2.4/topics/bf.test